Solutions and Colligative Properties

1. Solution $\rightleftharpoons$ Solute + Solvent

Factors Affecting Solubility: a. Nature of sol^n

b. Temp. generally $\Delta H > 0 \Rightarrow T\uparrow S\downarrow$

(* $Ce_2(SO_4)_3 \rightarrow \Delta H < 0$)

c. Pressure

2. Henry's Law: $P_{gas} = K_H \cdot \chi_{gas}$ → Solubility

$T\uparrow \Rightarrow K_H \uparrow \Rightarrow S\downarrow$

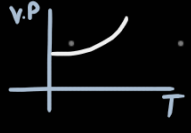
K_H: iski unit hamein solubility ki unit batayegi

χ_{gas}: in Liq.

3. Vapour Pressure: Pr. of vapour at equilibrium.

$P_{Solvent} > P_{Solution}$

(P_A°)



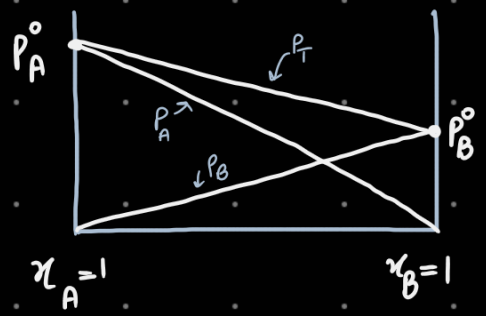
$$\log \frac{K_{P_2}}{K_{P_1}} = \frac{\Delta H}{2.303R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

4. Roult's Law: $P_A = P_A^\circ \chi_A$

$P_T = P_A + P_B$

$P_B = P_B^\circ \chi_B$

$(\chi_A + \chi_B = 1)$



5. χ_A - mole fraction in sol^n

y_A - mole fraction in vapour

$$P_A = y_A \times P_T = P_A^\circ \chi_A$$

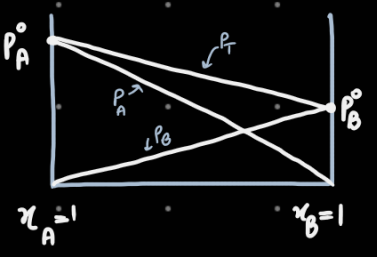
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6. Types of Soln

A. Ideal : Follows Raoult's law, $P_A = P_A^\circ x_A$

$$F_{A-A} = F_{A-B} = F_{B-B}$$

eg. Benzene + toulene
hexane + heptane
$C_2H_5Br + C_2H_5I$



$\Delta H_{soln} = 0$
$\Delta V_{mixing} = 0$
$\Delta G_{soln} < 0$
$\Delta S_{soln} > 0$

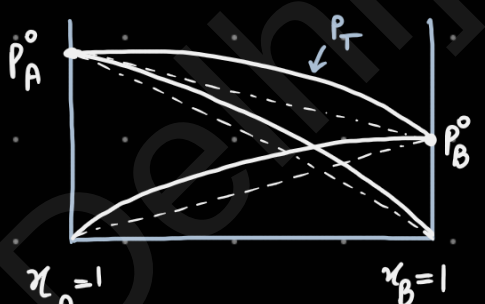
B. Non Ideal

(I) +ve deviation, $P_A > P_A^\circ x_A$

$$F_{A-A} > F_{A-B}$$
$$F_{B-B} > F_{A-B}$$

eg. $C_2H_5OH + H_2O$
$CH_3OH + H_2O$

(They show min^m B.P Azeotrope)



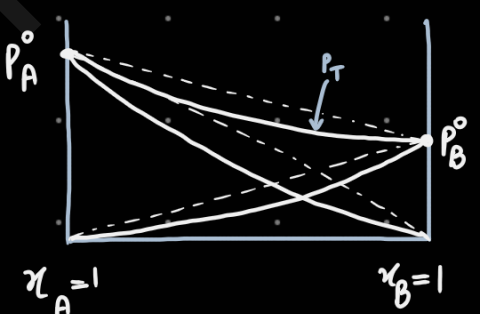
$\Delta H_{soln} > 0$
$\Delta V_{mixing} > 0$
$\Delta G_{soln} < 0$
$\Delta S_{soln} > 0$

(II) −ve deviation, $P_A < P_A^\circ x_A$

$$F_{A-A} < F_{A-B}$$
$$F_{B-B} < F_{A-B}$$

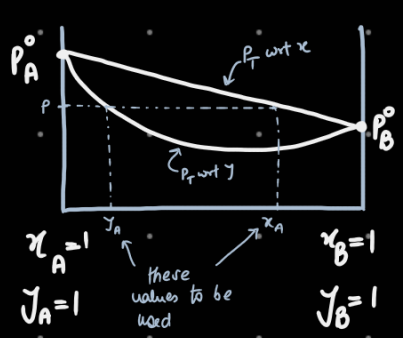
eg. $HNO_3 + H_2O$
$HX + H_2O$
$CH_3 - \underset{\underset{H}{|}}{\overset{\overset{H}{|}}{C}} - CH_3$

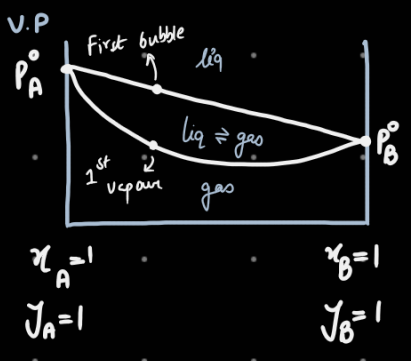
(They show max^m B.P Azeotrope)

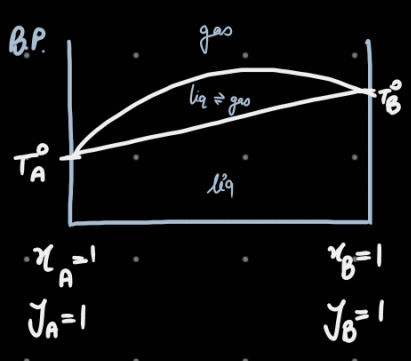


$\Delta H_{soln} < 0$
$\Delta V_{mixing} < 0$
$\Delta G_{soln} < 0$
$\Delta S_{soln} > 0$

7. For a Soln :







$P_A^\circ > P_B^\circ \quad \therefore T_A^\circ < T_B^\circ$

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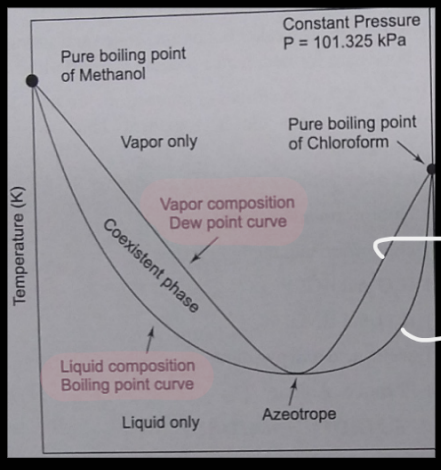
8. Azeotrope $x_A = y_A$

① Min^m B.P Azeotrope

eg. 95% ethanol + 5% H_2O

↳ 100°C

80°C → 78.2°C

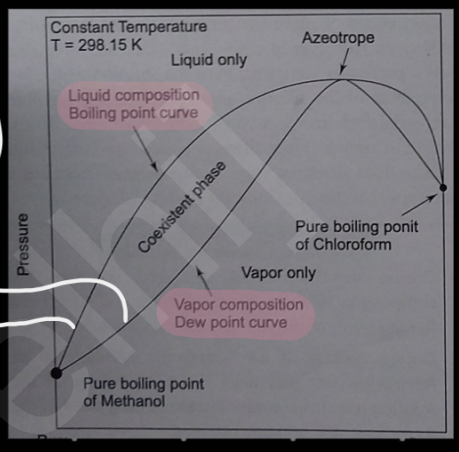


(Non-Azeotropes mein ye graphs touch XXX)

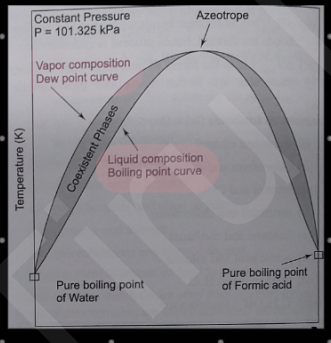
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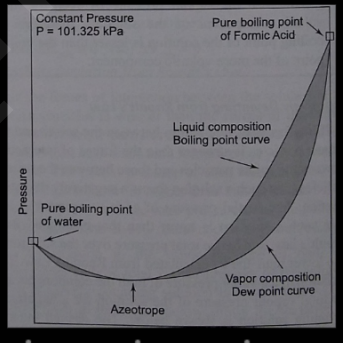
Vapour ka composition

Liquid ka composition



② Max^m B.P Azeotrope





eg. 68% HNO_3 + H_2O

86°C → 120°C ← 100°C

9. Colligative Properties

i. Relative Lowering in VP : $\frac{P_A^\circ - P_A}{P_A^\circ} = x_B$

ii. Elevation in BP : $\Delta T_b = K_b \, m$ (m ↳ molality)

Ebullioscopic const. → $K_b = \frac{MM_{solvent} \times R \times (T_b^\circ)^2}{1000 \times \Delta H_{vap}}$

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iii. Depression in freezing pt. : $\Delta T_f = K_f m$

Cryoscopic const. $\left(K_f = \frac{MM \times R \times (T_f^\circ)^2}{1000 \times \Delta H_{fusion}}\right)$

iv. Osmotic Pressure : $\pi = cRT$ → Temp, ↓ Conc^n, ↳ Gas const.

isotonic → $\pi_1 = \pi_2$ Hypertonic ↗↘ $\pi_1 > \pi_2$ ↖↙ Hypotonic

10. Van't Hoff factor (i) : $\frac{\text{Total no. of particles after diss/Ass.}}{\text{" " " before "}}$ $\left\{ i = \frac{(C.P)_{obs}}{(C.P)_{theoritical}} \right\}$ eg. $\Delta V.P = i\chi$, $\Delta T_b = i K_b m$

eg. $NaCl \rightleftharpoons Na^+ + Cl^-$ (C; C − Cα; Cα; Cα)

α ↳ degree of diss

$i = \frac{C - C\alpha + C\alpha + C\alpha}{C} = 1 + \alpha$

eg. $nA \rightleftharpoons A_n$ (C; C − Cα; $\frac{C\alpha}{n}$)

$i = 1 - \frac{(n-1)\alpha}{n}$

α ↳ degree of association

* CH_3COOH → aq. → dissociates
  ↳ benzene → associates

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